

An Evaluation of Non-Intrusive Full Body Tracking for CAVEs

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ABSTRACT

This paper proposes and evaluates non-intrusive full body tracking (FBT) for a CAVE. The non-intrusive FBT system is based on 3D Human Pose Estimation (HPE) with a single input device. The non-intrusive concept aims to enhance user experience through the lack of external trackers or devices the user is usually required to wear for FBT as well as place the tracking sensor behind the user. For the evaluation a proof-of-concept was created and a user study performed (N=42) to directly compare non-intrusive FBT with a standard input device. Our findings highlight the potential and challenges of non-intrusive FBT.

Index Terms: Virtual Reality, Full Body Tracking, Natural Interactions, Human Pose Estimation, User Study

1 INTRODUCTION

With recent technological developments in the field of Virtual Reality (VR), the interest in natural interactions has also grown [3, 8]. One form of natural interaction is full body tracking (FBT). However, a majority of that research focuses on FBT for head-mounted displays (HMDs). Next to HMDs, there are also large-scale immersive installations. These installations range from powerwalls, display corners, to CAVEs [11].

This paper proposes and evaluates non-intrusive FBT for a CAVE. The idea of non-intrusiveness aims to reduce distractions and let the user focus on the experience by not requiring them to wear any additional external devices. This proposal is based on Human Pose Estimation (HPE) and only requires a single RGB camera as input for FBT. The paper's contributions are as follows: (1) A proposal of a non-intrusive FBT system for a CAVE. (2) An evaluation of the proposal through a within-subjects study comparing a common input device and FBT (N=42).

2 CONCEPT

Our motivation is to examine the benefits of non-intrusive FBT for CAVEs. Therefore we created a proof-of-concept (PoC) and directly compared a common input device (CI) with FBT.

The output hardware for the PoC is a five-sided CAVE system [10]. A Google Pixel [5] streamed a live RGB video for FBT. So, we were wireless and flexible with the placement of the sensor. We used an ART Flystick [2] for CI.

As for the software, we used MediaPipe a 3D HPE algorithm [9]. A custom python script was used to process the live data of MediaPipe. The script defines the FBT settings for fast tracking required for real-time VR. For CI, we used the ART system [2], which can supply the data at 180hz, depending on the system configuration.

We created an interactive immersive experience called *CAVEknight* (see Figure 1) using Unreal Engine 5.4 [1] for a CAVE system that could combine both FBT and CI. This application requires users to dodge flying objects with their bodies. Additionally, the users can slice certain objects using a virtual



Figure 1: Experiment: User in *CAVEknight* using FBT with virtual sword (right) and projectiles (center).

sword, creating a challenge to dodge and collect simultaneously. The users start with five lives, if they are hit by certain objects, they lose a life. If they lose all their lives, the game ends.

3 METHODOLOGY

We conducted a user study to investigate the usability, benefits and challenges of FBT in a CAVE in comparison to CI. A within-subject user study was conducted in a controlled lab facility with a low number of distractions. Participants were guided through the study by a study supervisor. The study design was approved by the ethics commission of the authors institutions. Three main empirical elements are included:

Experiment: The user study consisted of the participant playing *CAVEknight* in a CAVE with the respective tracking methods of FBT and CI. The order of the input methods was alternated for the participant group. For FBT, the RGB sensor was placed outside the CAVE to not distract the participants and be non-intrusive.

Questionnaire: Quantitative feedback was collected from participants on their perception of the input methods through a questionnaire. The participants alternated between the questionnaire and experiment throughout the study. For their perception three dimensions are used: *perceived workload* through NASA-TLX [6, 12], *perceived user experience* through the UEQ [7] and the *users' sense of presence* with the IPQ [14]. A comment box was added to collect general feedback at the end of the questionnaire.

Semi-Structured Interview: An interview was conducted with each participant after the experiments and questionnaire to better understand their evaluation of the input methods as well as the *user awareness* of the tracking system. The following questions were asked: Which input method did you prefer, and can you explain why? What would you improve on the FBT input method? How aware were you of the tracking system?

4 EVALUATION AND DISCUSSION

Overall, 42 participants (22 female, 19 male, 1 diverse) took part in the study. Participants were recruited through flyers/posters and various university mailing lists. There were some participants who had no prior experience in VR or FBT. The participant group had a little more than average VR experience ($M=2.857$, $SD=1.241$) with 1=no experience and 5=expert level of experience. 17% of participants classified their level of VR experience as expert.

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Table 1: Evaluation: Paired samples t-test results for the individual NASA-TLX scores comparing FBT to CI (significant results in bold).

Item	FBT		CI		<i>t</i>	<i>df</i>	<i>p</i>	<i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>				
Mental Demand	4.881	1.418	3.738	1.415	4.716	41	<.001	0.728
Physical Demand	5.429	1.516	4.048	1.324	5.733	41	<.001	0.885
Temporal Demand	3.786	1.881	3.024	1.554	2.821	41	0.007	-
Performance	3.143	1.441	4.048	1.513	-2.917	41	0.006	-
Effort	4.976	1.370	3.738	1.289	5.368	41	<.001	0.828
Frustration	3.762	1.722	4.429	1.727	-1.738	41	0.090	-

Table 2: Evaluation: Paired samples t-test for UEQ (overall and subscale) scores comparing FBT and CI.

Item	FBT		CI		<i>t</i>	<i>df</i>	<i>p</i>	<i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>				
UEQ	0.548	0.921	0.955	0.913	-2.028	41	0.049	-0.313
Pragmatic Quality	-0.423	1.353	1.214	1.179	-5.867	41	<.001	-0.905
Hedonic Quality	1.518	0.939	0.696	1.235	3.843	41	<.001	0.593

Perceived workload: multiple paired sample t-tests were conducted with the individual items of the perceived workload grouped by input method (FBT and CI; see Table 1). Regarding the individual items of perceived workload, participants perceived the various demands (mental and physical) and effort to be higher for FBT than CI, which is reflected in the significant difference (see Table 1). Within the individual subscales, the largest differences are between **mental activity, physical activity and effort**.

Perceived Usability: Paired sample t-tests were conducted for the perceived usability subcategories as well as the individual items, depending on the input method (FBT or CI; Table 2). Overall, the usability of FBT lies at 0.548, which indicates that the FBT system has a **below-average usability** based on the handbook [13]. Within the UEQ, the **hedonic quality** of FBT lies at **good** ($M=1.518$, $SD=0.939$), indicating that the FBT system elicits feelings of novelty and enjoyment. However, the **pragmatic quality** of the FBT system can be interpreted as **bad** ($M=-0.423$, $SD=1.353$), with a low usability score in terms of functionality. There is no significant difference within the overall UEQ score between FBT and CI but in the subscales (see Table 2).

Perceived Presence: There is no significant difference with perceived presence between FBT and CI.

Thematic analysis was used as an inductive method to identify the themes and topics in the qualitative data (comment box and the semi-structured interview) [4] in four iterations over 8 hours. The evaluation was performed by the study supervisor. Five themes were generated: *Feedback to FBT*, *Preference to CI*, *Improvement Suggestions to FBT*, *Advantages of FBT and Disadvantages of FBT*. The three themes are directly connected to *feedback to FBT* with a focus on *improvements*, *advantages* and *disadvantages* of the system and some reasons to why they *preferred CI*. The most frequently cited advantage of FBT revolved around participants feeling FBT was *more fun and enjoyable*. Regarding tracking system awareness, five participants claimed they were aware of the tracking system and sensor location for both systems. 69% of participants stated they were not aware of the tracking systems.

The majority of participants (61.9%) preferred CI over FBT and found the overall usability of the non-intrusive FBT to be below average. A major contribution to the low usability may be the current implementation.

5 CONCLUSION AND FUTURE WORK

This paper proposes and evaluates non-intrusive FBT for a CAVE. For this a within-subjects study was performed ($N=42$), where participants could try both the FBT system and directly compare their experience with a CI device for immersive experiences. While users reception was positive towards the non-intrusive tracking, the im-

plementation highlights the issue with HPEs (jitter). With the rapid developments in the field of HPE, we are confident in the non-intrusive FBT for immersive installations and VR in general.

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